

Topics: 12.4 The Cross Product; 12.5 Equations of Lines and Planes (Part I)

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1. Compute $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{u} \times \mathbf{v}$ for each of the following pairs of vectors:

(a) $\mathbf{u} = \langle -1, 1, 2 \rangle$, $\mathbf{v} = \langle 4, 5, -2 \rangle$

$$\mathbf{u} \cdot \mathbf{v} = \langle -1, 1, 2 \rangle \cdot \langle 4, 5, -2 \rangle = -4 + 5 - 4 = -3.$$

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 2 \\ 4 & 5 & -2 \end{vmatrix} = -12\mathbf{i} + 6\mathbf{j} - 9\mathbf{k}.$$

(b) $\mathbf{u} = \mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{v} = 2\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$

$$\mathbf{u} \cdot \mathbf{v} = (\mathbf{i} - \mathbf{j} + 3\mathbf{k}) \cdot (2\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}) = 2 + 2 + 18 = 22.$$

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 3 \\ 2 & -2 & 6 \end{vmatrix} = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}.$$

(c) $\mathbf{u} = \langle 1, 2, 3 \rangle$, $\mathbf{v} = \langle -3, 0, 1 \rangle$

$$\mathbf{u} \cdot \mathbf{v} = \langle 1, 2, 3 \rangle \cdot \langle -3, 0, 1 \rangle = -3 + 0 + 3 = 0.$$

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ -3 & 0 & 1 \end{vmatrix} = 2\mathbf{i} - 10\mathbf{j} + 6\mathbf{k}.$$

- (d) What is the significance of your answers to parts (b) and (c)?

For part (b), the cross product is the zero vector so the vectors are parallel. Indeed,

$$\langle 2, -2, 6 \rangle = 2\langle 1, -1, 3 \rangle.$$

For part (c), the dot product is zero, so the vectors are orthogonal.

2. Consider $\mathbf{a} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $\mathbf{b} = -2\mathbf{j} + \mathbf{k}$.

- (a) Find the cross product $\mathbf{a} \times \mathbf{b}$.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -3 \\ 0 & -2 & 1 \end{vmatrix} = (1 - (6))\mathbf{i} - (2 - (0))\mathbf{j} + (-4 - (0))\mathbf{k} = -5\mathbf{i} - 2\mathbf{j} - 4\mathbf{k} = \langle -5, -2, -4 \rangle$$

- (b) Verify that $\mathbf{a} \times \mathbf{b}$ is orthogonal to $\mathbf{a}$.

We calculate that

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = \langle -5, -2, -4 \rangle \cdot \langle 2, 1, -3 \rangle = -10 - 2 + 12 = 0.$$

Thus $\mathbf{a} \times \mathbf{b}$ is orthogonal to $\mathbf{a}$.

- (c) Find two unit vectors orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

The cross product of two vectors is orthogonal to both vectors. From part (a), $\mathbf{a} \times \mathbf{b} = \langle -5, -2, -4 \rangle$. Taking the magnitude,

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{(-5)^2 + (-2)^2 + (-4)^2} = \sqrt{45} = 3\sqrt{5}.$$

Thus, two unit vectors orthogonal to both $\mathbf{a}$ and $\mathbf{b}$ are

$$\left\langle -\frac{5}{3\sqrt{5}}, -\frac{2}{3\sqrt{5}}, -\frac{4}{3\sqrt{5}} \right\rangle \quad \text{and} \quad \left\langle \frac{5}{3\sqrt{5}}, \frac{2}{3\sqrt{5}}, \frac{4}{3\sqrt{5}} \right\rangle$$

3. Consider the points $P(1, 2, 1)$, $Q(2, 5, 4)$, $R(6, 9, 12)$, and $S(5, 6, 9)$ in $\mathbb{R}^3$.

- (a) Show that the vectors $\overrightarrow{PQ}$, $\overrightarrow{PR}$, and $\overrightarrow{PS}$ are coplanar.

We have $\overrightarrow{PQ} = \langle 1, 3, 3 \rangle$, $\overrightarrow{PR} = \langle 5, 7, 11 \rangle$, and $\overrightarrow{PS} = \langle 4, 4, 8 \rangle$. Then,

$$\overrightarrow{PQ} \cdot (\overrightarrow{PR} \times \overrightarrow{PS}) = \begin{vmatrix} 1 & 3 & 3 \\ 5 & 7 & 11 \\ 4 & 4 & 8 \end{vmatrix} = 1(56 - 44) - 3(40 - 44) + 3(20 - 28) = 12 + 12 - 24 = 0.$$

Since this scalar triple product is zero, the volume of the parallelepiped determined by the three vectors is zero. Therefore, the vectors are coplanar.

- (b) Find the area of the parallelogram $PQRS$.

The parallelogram is determined by the vectors $\overrightarrow{PQ} = \langle 1, 3, 3 \rangle$ and $\overrightarrow{PS} = \langle 4, 4, 8 \rangle$. Their cross product is

$$\overrightarrow{PQ} \times \overrightarrow{PS} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & 3 \\ 4 & 4 & 8 \end{vmatrix} = (24 - 12)\mathbf{i} - (8 - 12)\mathbf{j} + (4 - 12)\mathbf{k} = 12\mathbf{i} + 4\mathbf{j} - 8\mathbf{k}.$$

So, the area of parallelogram $PQRS$ is

$$|\overrightarrow{PQ} \times \overrightarrow{PS}| = \sqrt{12^2 + 4^2 + (-8)^2} = 4\sqrt{14}.$$

- (c) Find the area of triangle PQS .

The area of triangle PQS is half the area of the parallelogram $PQRS$. Using part (b), since the area of the parallelogram is $4\sqrt{14}$, the area of triangle PQS is therefore $2\sqrt{14}$.

4. Find the following equations of the line through the point $P(2, 2, 4)$ that is parallel to the vector $\mathbf{a} = 4\mathbf{i} - \mathbf{j} + 7\mathbf{k}$. For parts (a) and (b), use the parameter t .
- (a) Vector equation
 - (b) Parametric equations
 - (c) Symmetric equations

The position vector of the point P is $\overrightarrow{OP} = 2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Since the line is parallel to $\mathbf{a}$, we may take the direction vector to be $\mathbf{v} = \mathbf{a} = 4\mathbf{i} - \mathbf{j} + 7\mathbf{k}$. Any point on the line has position vector $\mathbf{r}$.

(a) A vector equation of the line through P with direction vector $\mathbf{v}$ is

$$\mathbf{r}(t) = \overrightarrow{OP} + t\mathbf{v} = (2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}) + t(4\mathbf{i} - \mathbf{j} + 7\mathbf{k}).$$

Equivalently,

$$\mathbf{r}(t) = (2 + 4t)\mathbf{i} + (2 - t)\mathbf{j} + (4 + 7t)\mathbf{k}.$$

(b) Parametric equations are

$$x(t) = 2 + 4t, \quad y(t) = 2 - t, \quad z(t) = 4 + 7t.$$

(c) Symmetric equations are

$$\frac{x - 2}{4} = \frac{y - 2}{-1} = \frac{z - 4}{7}.$$

5. (a) Find symmetric equations of the line through the point $P(3, 4, 0)$ that is perpendicular to both vectors $2\mathbf{i} + 2\mathbf{j}$ and $\mathbf{j} + \mathbf{k}$.

A direction vector $\mathbf{v}$ for the desired line must be perpendicular to both given vectors, so we can take the cross product:

$$\mathbf{v} = (2\mathbf{i} + 2\mathbf{j}) \times (\mathbf{j} + \mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}.$$

Thus a direction vector is $\langle 2, -2, 2 \rangle$. The symmetric equations are

$$\frac{x - 3}{2} = \frac{y - 4}{-2} = \frac{z - 0}{2}.$$

- (b) Find the point where the line from part (a) intersects the yz -plane.

The yz -plane is given by $x = 0$. Using the symmetric equations,

$$\frac{0 - 3}{2} = \frac{y - 4}{-2} = \frac{z - 0}{2}.$$

So

$$-\frac{3}{2} = \frac{y - 4}{-2} \quad \text{and} \quad -\frac{3}{2} = \frac{z}{2}.$$

Solving gives $y = 7$ and $z = -3$. Therefore, the line intersects the yz -plane at

$$(0, 7, -3).$$

6. Determine whether the lines L_1 and L_2 are parallel, intersecting, or skew.

(a) $L_1 : \mathbf{r}(t) = \langle -1 + 3t, 2 + 4t, 3 - 2t \rangle$, $L_2 : \frac{x-1}{2} = \frac{y}{-3} = \frac{z+1}{-3}$

Write L_2 in parametric form by letting

$$\frac{x-1}{2} = \frac{y}{-3} = \frac{z+1}{-3} = s.$$

Then

$$x = 1 + 2s, \quad y = -3s, \quad z = -1 - 3s,$$

so a direction vector for L_2 is $\mathbf{v}_2 = \langle 2, -3, -3 \rangle$. A direction vector for L_1 is $\mathbf{v}_1 = \langle 3, 4, -2 \rangle$.

Since $\mathbf{v}_1$ is not a scalar multiple of $\mathbf{v}_2$, the lines are not parallel. To check whether they intersect, solve

$$-1 + 3t = 1 + 2s, \quad 2 + 4t = -3s, \quad 3 - 2t = -1 - 3s.$$

From the first two equations, we obtain $t = \frac{2}{17}$ and $s = -\frac{14}{17}$. Substituting into the third equation gives

$$3 - 2\left(\frac{2}{17}\right) \neq -1 - 3\left(-\frac{14}{17}\right),$$

so the system is inconsistent and the lines do not intersect. Therefore, the lines are **skew**.

(b) $L_1 : x = 2t, y = -3 + t, z = 5 - t$, $L_2 : x = 3 - 3s, y = 2 - \frac{3}{2}s, z = \frac{3}{2}s$

Direction vectors are $\mathbf{v}_1 = \langle 2, 1, -1 \rangle$ and $\mathbf{v}_2 = \langle -3, -\frac{3}{2}, \frac{3}{2} \rangle$. Since

$$\mathbf{v}_1 = -\frac{2}{3}\mathbf{v}_2,$$

the lines are **parallel**. Note that they are not the same line (for example, $(0, -3, 5)$ lies on L_1 when $t = 0$, but it does not satisfy the equations of L_2 for any s).

7. Find an equation of the plane through the points $A(0, 1, 2)$, $B(1, 2, 3)$, and $C(2, 3, 5)$.

First find direction vectors in the plane:

$$\overrightarrow{AB} = \langle 1, 1, 1 \rangle, \quad \overrightarrow{AC} = \langle 2, 2, 3 \rangle.$$

A normal vector is their cross product:

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 2 & 2 & 3 \end{vmatrix} = (1 \cdot 3 - 1 \cdot 2)\mathbf{i} - (1 \cdot 3 - 1 \cdot 2)\mathbf{j} + (1 \cdot 2 - 1 \cdot 2)\mathbf{k} = \langle 1, -1, 0 \rangle.$$

Using point $A(0, 1, 2)$, the plane equation is

$$1(x - 0) + (-1)(y - 1) + 0(z - 2) = 0.$$

This can be rewritten as:

$$x - y = -1.$$

8. Find an equation of the plane through the points $(0, -2, 5)$ and $(-1, 3, 1)$ that is perpendicular to the plane $2z = 5x + 4y$.

Rewrite the given plane as $5x + 4y - 2z = 0$, so a normal vector is $\mathbf{n}_1 = \langle 5, 4, -2 \rangle$.

Let $A(0, -2, 5)$ and $B(-1, 3, 1)$. Then

$$\overrightarrow{AB} = \langle -1, 5, -4 \rangle$$

lies in the desired plane, so it is perpendicular to the desired plane's normal vector $\mathbf{n}_2$. Also, since the planes are perpendicular, $\mathbf{n}_2$ is perpendicular to $\mathbf{n}_1$. Hence $\mathbf{n}_2$ is perpendicular to both $\overrightarrow{AB}$ and $\mathbf{n}_1$, so we can take

$$\mathbf{n}_2 = \overrightarrow{AB} \times \mathbf{n}_1 = \langle -1, 5, -4 \rangle \times \langle 5, 4, -2 \rangle = \langle 6, -22, -29 \rangle.$$

Using point $A(0, -2, 5)$, the plane equation is

$$6(x - 0) - 22(y + 2) - 29(z - 5) = 0.$$

This can be rewritten as:

$$6x - 22y - 29z = -101.$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Cross Product If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, then

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta.$$

Equations of a line. The line through the point (x_0, y_0, z_0) and parallel to the vector $\langle a, b, c \rangle$ has **vector equation**

$$\mathbf{r}(t) = \langle x_0, y_0, z_0 \rangle + t \langle a, b, c \rangle.$$

The **parametric equations** for this line are

$$x = x_0 + at \quad y = y_0 + bt \quad z = z_0 + ct.$$

We can eliminate t to obtain the **symmetric equations**

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c},$$

provided a , b , and c are all nonzero. If a component of the direction vector is zero, the corresponding coordinate is constant.

Equations of a plane. The plane through (x_0, y_0, z_0) with normal vector $\langle a, b, c \rangle$ is given by

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

By collecting terms, we can rewrite this equation in the form

$$ax + by + cz = d$$

where $d = ax_0 + by_0 + cz_0$ is a constant. Two planes are parallel if their normal vectors are parallel, and they are perpendicular if their normal vectors are orthogonal.

Suggested Textbook Problems

Section 12.4: 1-22, 27-38, 43-46

Section 12.5: 1-72